Indraprastha WORLD SCHOOL
Paschim Vihar, New Delhi- 110063
Session: 2024-25
Sample Paper
Subject: Mathematics

Class: VII **Maximum Marks: 80**
Time allowed: 3 hours

General instructions

- This question paper has 21 questions and 3 printed pages.
- All questions are compulsory to attempt.
- Follow the prescribed word limit while answering the questions.
- Mention the correct serial number of the question and attempt the paper neatly.

Section-A

1. Answer the following.

a. Two angles form a linear pair. If measure of one angle is of 46° then find the measure of other angle. **1**

b. Write the algebraic expression for : The sum of the numbers a and b subtracted from their product. **1**

c. What is the other name given to mean? **1**

d. Find the sum of 3m + 7 and -6m + 11. **1**

e. -7 + (-6) – 5 = _____ **1**

2. Fill in the following blanks.

a. Two lines are _____ when the pair of alternate angles are equal. **1**

b. If in a mushroom, $\frac{19}{20}$ is water, the fraction of solid is _____. **1**

c. 8 X $\frac{1}{120}$ is _____ than $\frac{1}{3}$. **1**

d. The product of -12, 3 and 10 is _____. **1**

e. The _____ is the observation that occurs most often in the given data. **1**

3. Choose the correct option from the following.

a. The integer 17 less than -25 is_____. **1**
i. 42 ii. -42 iii 8 iv. -8

b. Which of the following sum is not negative? **1**
i. -40 + (-20) ii. -70 + 55 iii. -73 + 92 iv -62 + 30

c. The value of $\frac{7}{3} + \frac{9}{2}$ is_____. **1**
i. $\frac{41}{6}$ ii. $\frac{16}{5}$ iii. $\frac{16}{6}$ iv. 12

d. $\frac{1}{5}$ of a non leap year is _____. **1**
i. 70 ii. 73 iii. 1825 iv. 300

e. The central value of the data is called _____. **1**
i. mean ii. median iii. mode iv. range

4. State whether the following statement is true or false. **2**

a. The sum of two -ve integers is always -ve.

b. Mean is the highest occurring observation in the data.

5. In the questions given below, there are two statements marked as Assertion(A) and Reasoning (R). Read the statements and choose the appropriate option. **3**

a. **Assertion:** If an angle formed by two intersecting lines is 60∘, then its vertically opposite angle is 60∘.
Reason: If two lines intersect each other, then the vertically opposite angles are equal.

i. Both assertion and reason are correct and reason is the correct explanation for assertion.
ii. Both assertion and reason are correct and reason is not the correct explanation for assertion.
iii. Assertion is correct and reason is wrong.
iv. Assertion is wrong and reason is correct.

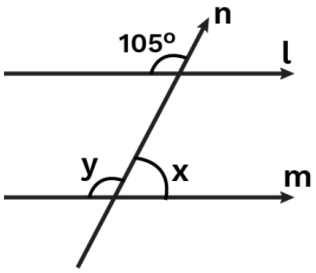
b. **Assertion:** The expression y+ 19 is in one variable.
Reason: In expression y+19, the variable is y.
i. Both assertion and reason are correct and reason is the correct explanation for assertion.
ii. Both assertion and reason are correct and reason is not the correct explanation for assertion.
iii. Assertion is correct and reason is wrong.
iv. Assertion is wrong and reason is correct.

c. **Assertion:** The mean of the numbers 10,20, 30 and 40 is 25.
Reason: Mean = 10+20+30+40/4 = 100/4 = 25.
i. Both assertion and reason are correct and reason is the correct explanation for assertion.
ii. Both assertion and reason are correct and reason is not the correct explanation for assertion.
iii. Assertion is correct and reason is wrong.
iv. Assertion is wrong and reason is correct.

Section-B

6. -100 + 25 – (-34) + _____ = -100. **2**

7. If lines l and m are parallel to each other then find the value of x and y. **2**



8. Solve (-3) + (-8)÷ (- 4) - 2 x (-2) **2**

9. Students of class VII scored the following marks in aa test: 36, 35, 50, 46, 60, 55 **2**
Calculate the mean of the marks scored in the test.

10. Find the average of first five whole numbers. Also find the range of the given data:
22, 16, 9, 15, 4.5, 8, 5. **2**

Section-C

11. The weight of a baby elephant was 218.99 kg. After two years, his weight increased by 109.85 kg. Find the weight of elephant after two years. **3**

12. Identify the supplementary and complementary angles. **3**
a. 60∘,120∘
b. 30∘,60∘
c. 35∘,145∘

13. The runs scored in a cricket match by 11 players is as follows- 6, 15, 120, 50, 100, 80, 10, 15, 8, 10 and 14. Find the range, median and mode of this data. **3**

14. Solve- { (-36) ÷ 4 } + {48 ÷ (-12)}. **3**

15. Aanya took a variable y and added 6 to four times of it. **3**
a) Write the algebraic expression for Aanya.
b) What is value of expression when y is equal to 4?

16. Frame a monomial, binomial and trinomial expression. **3**

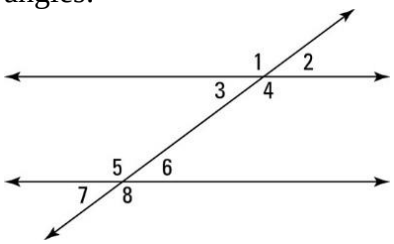
Section-D

17. Write five pair of integers (m, n) such that m ÷ n = -3. **5**

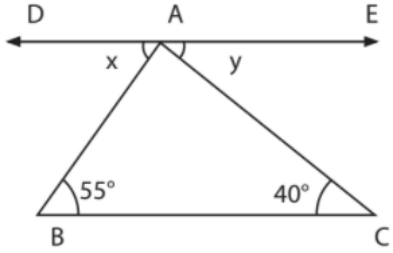
18. Consider the data collected from a survey in a school. Draw a double bar graph choosing an appropriate scale. **5**

Favourite music	Rock	Classic	Pop
Number of boys	10	8	14
Number of girls	12	10	6

19. If two parallel lines are cut by a transversal and measure of ∠1 is 125◦ then find the measure of all the angles: **5**



20. In the given figure, DE ∥ BC. Find the values of x and y. **5**



Section-E

21. Case study questions:

i. Three brothers were told to distribute a cake amongst themselves. The eldest brother ate 1/3 of the cake and the second brother ate 1/4 of the cake. Remaining cake was eaten by the youngest brother Based on this observation, answer the following questions.



a) What part of the cake was eaten by the youngest brother? **2**
b) Who ate the maximum part of the cake? **2**

ii. Sanya and Arpita decided to play a game. Their scores are given in the table below.

ROUNDS	Sanya	Arpita
I	-10	15
II	12	-20
III	15	16
IV	-9	-18



a) Whose performance is better. **2**
b) Who won in round IV? **2**

iii. Arun, Varun and Taruni are very intelligent students of their class. They involve the study in their games. In one of their games, they asked each other to write an algebraic expression to apply the arithmetic operations on those expressions. Arun, Varun and Taruni wrote the expressions (-3x + 5), (x - 7) and (3y + 1) respectively.



a) Find the co-efficient of 'x' in the expression written by Arun. **2**
b) Find the sum of the expressions written by Taruni and Varun. **2**